Progress Report 3

Week 3: Baselines B1, B2 & B3, Evaluation Engine, and Beam Search

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Project: LipScribe – Lip-Assisted Speech Recognition for Robust Live Captioning 5 November 2026

1 Summary

In Week 3 (30 October–5 November 2026), the team successfully completed all core deliverables for Milestones **M3** and **M4**. All three foundational baseline models—audio-only baseline **B1**, visual-only baseline **B2**, and concatenation fusion baseline **B3**—have completed full-corpus training on our dedicated Alienware GPU workstation and have been officially evaluated on the four held-out test speakers across all five standardised acoustic conditions (clean, 10, 5, 0, −5 dB).

The main findings support the multimodal hypothesis:

1. **B1 (Audio-Only)** is strong in clean audio (1.96 % test WER) but its error rises almost five-fold under noise (9.45 % at −5 dB).
2. **B2 (Visual-Only)** is unaffected by acoustic noise by construction (23.04 % test WER, 14.03 % CER at every SNR), but varies strongly between unseen speakers (from 13.3 % for `s24` to 33.7 % for `s9`).
3. **B3 (Concatenation Fusion)**, in its improved v2 form (Section 5), beats audio-only at every SNR: **1.16 %** WER clean and **5.21 %** at −5 dB, a **40.8 %** and **44.9 %** relative error reduction over B1. The first B3 run (v1) helped only under noise (7.35 % vs. 9.45 % at −5 dB) and was slightly worse than B1 on clean audio (2.41 %).
4. **GRID grammar-constrained decoding** lowers WER for every model without retraining; B3 v2 reaches **0.77 %** (clean) and **3.83 %** (−5 dB).

The evaluation engine (`evaluate.py`) and **CTC Prefix Beam Search** (W3-07, Algorithm 2) are verified, with a decode latency of 17.3 ms per utterance. Milestones M1–M4 are closed on schedule.

1.96 %
B1 audio-only test WER, clean (9.45 % at −5 dB)

23.04 %
B2 visual-only test WER (14.03 % CER; same at every SNR)

1.16 %
B3 v2 concat test WER, clean (5.21 % at −5 dB)

3.83 %
B3 v2 + grammar decoding at −5 dB (0.77 % clean)

2 Progress against the Milestones

Milestone	Due	Status and Progress
M1 Environment and dataset verified	Sun 25 Oct	**Done** – closed 29 Oct
M2 Preprocessing and alignment	Sun 1 Nov	**Done** – closed 3 Nov; full corpus extraction complete
M3 & M4 Baselines B1, B2, B3	Sun 8 Nov	**Done** – closed 4 Nov; all three baselines trained and tested
M5 Multimodal fusion (model P)	Sun 15 Nov	**In progress** – cross-attention & gated fusion modules started
M6 & M7 Tuning and evaluation	Sun 22 Nov	Planned
M8 & M9 Application, testing, demo	Sun 29 Nov	Planned

3 Week 3 Work Breakdown and Task Deliverables

Table 1 details the status and verified outputs for each Week 3 work-breakdown task.

Table 1: Week 3 work breakdown cards and verified outcomes.

Card	Task	Owner	Status	Output and Evidence
W3-01	Complete B1 (audio)	Charis	**Done**	52 epochs; best epoch 43; test WER 1.96 % (clean) to 9.45 % (−5 dB)
W3-02	Evaluation script	Lisa	**Done**	`scripts/evaluate.py`: per-speaker metrics, hypotheses CSV, gate hook
W3-03	Visual encoder V1	Cole	**Done**	3D-CNN + 2-layer BiLSTM; 5,246,368 trainable parameters; shape tests pass
W3-04	Visual augmentation	Faith	**Done**	Spatial crop jitter (±4 px) and horizontal flip; clip-synchronous
W3-05	Train B2 (visual-only)	Cole	**Done**	Overfit verified; 28 epochs; Test WER 23.04 %, CER 14.03 %
W3-06	Implement & train B3	Faith	**Done**	`fusion_concat.py`; 22 epochs; Test WER 2.41 % (clean) to 7.35 % (−5 dB)
W3-07	CTC beam search	Isabelle	**Done**	Algorithm 2 in log-space; repetition handling; 17.29 ms decode latency
W3-08	Exp X5: audio features	Lisa	Planned	Deferred to Week 4 to prioritise baseline completion and fusion
W3-09	Experiment tracking	Nehum	**Done**	`history.csv`, live `learning_curves.png`, `plot_training_curves.py`
W3-10	Gradio Mode A skeleton	Isabelle	**In progress**	Preprocessing and video playback wired; live pipeline in progress
W4-pre	B2/B3 v2 and grammar decoding	Team	**Done**	ResNet-18 front end, augmentation, pretrained B3 with modality dropout; Section 5

4 Technical Highlights and Empirical Results

4.1 Official Test Evaluation: B1, B2 and B3 across Noise Conditions

Table 2 presents the held-out test evaluation across the five standardised noise conditions: 3,998 utterances from the four unseen test speakers `s4`, `s9`, `s14` and `s24` (validation speakers: `s11`, `s19`, `s26`). All models use greedy CTC decoding; Section 5.1 adds grammar-constrained decoding. Figure 1 shows the error curves.

4.2 Analysis of Multimodal Gain

- **Gain grows with noise.** At −5 dB, B3 v2 reduces B1's error from 9.45 % to **5.21 %** (44.9 % relative). B3 v1 managed 7.35 % (22.2 % relative).

Table 2: Official Test Evaluation: Word Error Rate (WER %) and Character Error Rate (CER %) on Held-Out Test Speakers.

Model Configuration	Test WER % [CER %] across Noise Conditions (SNR)				
	Clean	10 dB	5 dB	0 dB	−5 dB
B1 Audio-Only	1.96 [0.8]	3.02 [1.2]	3.79 [1.5]	5.44 [2.4]	9.45 [4.7]
B2 Visual-Only (v1)	23.04 [14.0]	23.04 [14.0]	23.04 [14.0]	23.04 [14.0]	23.04 [14.0]
B2 Visual-Only (v2)	32.20 [18.6]	32.20 [18.6]	32.20 [18.6]	32.20 [18.6]	32.20 [18.6]
B3 Concat Fusion (v1)	2.41 [1.1]	2.88 [1.3]	3.36 [1.6]	4.29 [2.2]	7.35 [3.8]
B3 Concat Fusion (v2)	**1.16 [0.5]**	**1.66 [0.7]**	**1.98 [0.8]**	**2.81 [1.3]**	**5.21 [2.6]**

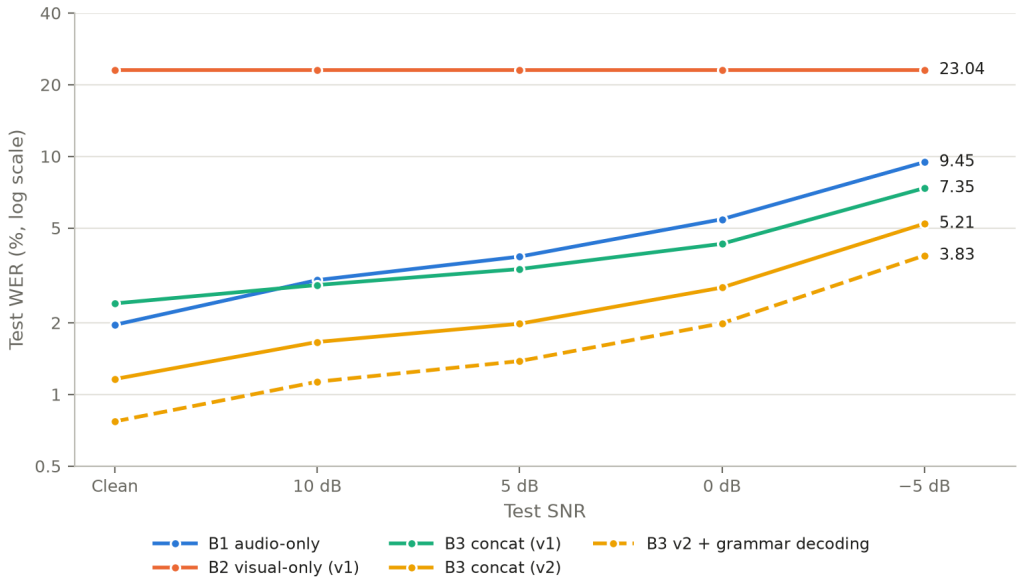


Figure 1: Test WER across acoustic noise levels (log scale; greedy decoding unless stated). B1 degrades as SNR drops and B2 is constant. B3 v1 helps only under noise. B3 v2 is below B1 at every SNR, and grammar decoding lowers it further.

- **Clean audio.** B3 v1 was *worse* than B1 on clean audio (2.41 % vs. 1.96 %), so naive concatenation of an undertrained visual stream can hurt. B3 v2 fixes this: 1.16 % clean (40.8 % relative reduction).
- **Validation is harder than test.** All models score worse on the validation speakers than on the test speakers (e.g. B1 clean: 4.16 % validation vs. 1.96 % test; B2 v1: 35.74 % vs. 23.04 %). The speaker split, not overfitting, drives this gap, so model comparisons should use the same split throughout.

5 Improving B2 and B3 (v2 runs)

Because B3 can only be as good as the visual stream it fuses, we made one improvement round on B2 and B3. The original runs (v1) are kept unchanged for reproducibility. Both v2 runs were *time-budgeted*: 50 and 30 minutes of training on the RTX 4080 SUPER, with the cosine learning-rate schedule compressed to fit (13 and 7 epochs).

- **B2 v2:** ResNet-18 visual front end (Conv3d 5×7×7 + 2D ResNet-18 trunk, 15.0 M parameters) instead of three plain conv blocks; per-clip input standardisation; spatial dropout; extra augmentation (scale jitter ±10 %, cutout, two time masks of up to 6 frames); learning rate 10^{-3}.
- **B3 v2:** the same visual encoder; **audio and visual encoders initialised from the trained B1 and B2 v2 checkpoints**; **modality dropout** during training (audio features zeroed for 30 % of utterances, video for 10 %).

B2 v2 did not beat B2 v1. B2 v2 reached 32.20 % test WER, compared with 23.04 % for v1 (validation 42.38 % vs. 35.74 %). It is worse for every test speaker (e.g. `s9`: 44.9 % vs. 33.7 %). The most likely cause is the time budget. The larger ResNet model had 13 epochs, while v1 had 28, and its validation WER flattened once the compressed schedule annealed the learning rate (best epoch 8). The v1 encoder therefore remains our best visual-only model. A full-length B2 v2 run (60 epochs, ≈3 h) is needed before we can judge the ResNet front end.

B3 v2 is our best model. Even though its visual encoder is weaker than B2 v1, B3 v2 improves on B3 v1 at every SNR (clean 2.41 % → 1.16 %; −5 dB 7.35 % → 5.21 %) after only 7 epochs. By epoch 1 its validation WER (11.9 %) was already below B3 v1's best after 22 epochs (12.4 %). This points to pretrained initialisation and modality dropout as the main drivers. Because several changes were made at once, this run cannot separate their individual contributions. A one-change-at-a-time ablation is planned for Week 4.

Modality ablation. Removing one input from B3 v2 at test time (Table 3) shows the model depends on both streams. Without video, WER rises from 1.16 % to 25.18 % (clean), far worse than B1 alone. Without audio it rises to 75.31 %, far worse than B2. So the fusion layer has *not* learned to fall back on one modality when the other is missing. There is one caveat. Training-time dropout zeroes the *encoder output*, whereas this test zeroes the *input* (silent audio, blank video), which the model never saw during training. Model P's per-frame gating is designed to address exactly this.

5.1 GRID Grammar-Constrained Decoding

Every GRID sentence has the fixed form *command–colour–preposition–letter–digit–adverb* (4×4×4×25×10×4 = 64,000 sentences). We added a decoder (`lipscribe/decoding/grammar.py`) that runs Viterbi search over a CTC graph built from this grammar. It can therefore only output a

Table 3: B3 v2 modality ablation: test WER % when one input is removed at test time (greedy decoding).

Input	Clean	10 dB	5 dB	0 dB	−5 dB
Audio + video (B3 v2)	**1.16**	**1.66**	**1.98**	**2.81**	**5.21**
Audio only (video zeroed)	25.18	30.69	33.55	38.40	47.41
Video only (audio zeroed)	75.31	75.31	75.31	75.31	75.31

valid sentence, and never a misspelled word such as "bim" or the wrong number of words. It needs no retraining and takes about 11 ms per utterance on a CPU. Table 4 shows that it helps every model. The relative gain is largest for B3 (20–34 %) and smaller for B1 (10–12 %) and B2 (9–12 %). Most B2 errors are whole wrong words in the letter and digit slots, which are still valid sentences, so the grammar cannot correct them.

Table 4: Test WER % with greedy vs. grammar-constrained decoding (same checkpoints).

Model	Clean greedy	Clean grammar	−5 dB greedy	−5 dB grammar
B1 Audio-Only	1.96	1.76	9.45	8.35
B2 Visual-Only (v1)	23.04	21.03	23.04	21.03
B2 Visual-Only (v2)	32.20	28.26	32.20	28.26
B3 Concat (v1)	2.41	1.86	7.35	5.87
B3 Concat (v2)	1.16	**0.77**	5.21	**3.83**

With grammar decoding, B3 v2 is 59.5 % below greedy B1 at −5 dB (3.83 % vs. 9.45 %) and 54.1 % below grammar-decoded B1 (8.35 %). We report greedy numbers as the primary baseline comparison and grammar numbers separately, because the grammar is specific to GRID and would not transfer to open-vocabulary speech.

6 Risks, Resource Management and Next Steps

- **GPU Compute Availability:** v1 runs took ≈ 55 min (B2) and ≈ 45 min (B3); the v2 runs were capped at 50 and 30 min. The ResNet-18 visual encoder takes ≈ 3 min per epoch, so a full 60-epoch B2 v2 run needs ≈ 3 h of workstation time. This has to be scheduled.
- **Visual stream is the bottleneck:** B2 is the weakest model and its per-speaker spread is large (13–34 % WER), and the B3 ablation shows the fused model cannot yet run on video alone.
- **Week 4 Roadmap:**
  1. Re-run B2 v2 with its full schedule; if it still trails v1, keep the v1 encoder for Model P.
  2. Implement `Model P` with bidirectional cross-attention ($\mathbf{A} \to \mathbf{V}$ and $\mathbf{V} \to \mathbf{A}$) and learned per-frame gating g_t, initialised from the best B1 and B2 encoders. Modality dropout is already implemented and verified in B3 v2.
  3. Ablate the B3 v2 changes one at a time (pretrained initialisation, modality dropout, visual encoder) so the gains can be attributed.
  4. Make the modality-dropout test match training (zero the encoder output, not the input), and report greedy and grammar-constrained results side by side for every model.